

Patient name:	Bastian Iskandar	Sample type:	gDNA	Report date:	29-JUL-2023
DOB:	05-MAR-2010	Sample collection date:	11-JUL-2023	Invitae #:	RQ5340773
Sex assigned at birth:	Male	Sample accession date:	21-JUL-2023	Clinical team:	Cut Hafifah
Gender:					
Patient ID (MRN):	4719127				

Reason for testing

Diagnostic test for a personal history of disease

Test performed

Sequence analysis and deletion/duplication testing of the gene listed in the Genes Analyzed section.

- Invitae Neurofibromatosis Type 1 Test


RESULT: POSITIVE

One Pathogenic variant identified in NF1. NF1 is associated with autosomal dominant neurofibromatosis type 1, neurofibromatosis-Noonan syndrome and Watson syndrome.

GENE	VARIANT	ZYGOSITY	VARIANT CLASSIFICATION
NF1	c.5546+5G>A (Intronic)	heterozygous	PATHOGENIC

About this test

This diagnostic test evaluates 1 gene(s) for variants (genetic changes) that are associated with genetic disorders. Diagnostic genetic testing, when combined with family history and other medical results, may provide information to clarify individual risk, support a clinical diagnosis, and assist with the development of a personalized treatment and management strategy.

Clinical comments

- When a single Variant of Uncertain Significance is found in a requisitioned gene that is only associated with autosomal recessive condition(s), it may not be included in the report.



Next steps

- This is a medically important result that should be discussed with a healthcare provider, such as a genetic counselor, to learn more about this result and the appropriate next steps for further evaluation, treatment and/or management. This result should be interpreted within the context of additional laboratory results, family history and clinical findings.
- Please see NCCN (www.nccn.org), PMID: 30006586, PMID: 28620004, PMID: 31010905, PMID: 35657359, and PMID: 35788692 for management guidelines regarding NF1-related condition(s).
- Consider sharing this result with relatives as they may also be at risk. Details on our Family Variant Testing program can be found at www.invitae.com/family.
- Register your test at www.invitae.com/patients to download a digital copy of your results. You can also access educational resources about how your results can help inform your health.

Clinical summary

A Pathogenic variant, c.5546+5G>A (Intronic), was identified in NF1.

- The NF1 gene is associated with autosomal dominant neurofibromatosis type 1 (NF1) (MedGen UID: 18013), neurofibromatosis-Noonan syndrome (NFNS) (MedGen UID: 419089), and Watson syndrome (MedGen UID: 107817).
- This result is consistent with a predisposition to, or diagnosis of, NF1-related conditions.
- NF1 is a neurocutaneous disorder associated with an increased risk of several types of benign and malignant tumors. Primary features include multiple cafe-au-lait spots, inguinal freckling, and neurofibromas. NF1 is highly variable in its expression, even among affected members of the same family. There is an increased risk for central nervous system neoplasms and other malignancies including female breast cancer, gastrointestinal stromal tumors, pheochromocytoma, and soft tissue sarcomas (PMID: 20301288). Approximately one-half of NF1 cases are inherited, and the remainder are the result of spontaneous de novo mutations (PMID: 20082463).

Allelic NF1 disorders include Watson syndrome and NFNS. Watson syndrome is characterized by pulmonic stenosis, cafe-au-lait spots, macrocephaly, Lisch nodules, intellectual disability, and short stature (PMID: 1770531, 23047742). NFNS is characterized by classic neurofibromatosis symptoms in addition to short stature, distinctive facial features, broad neck, congenital heart defects, speech delay, and pectus anomaly (PMID: 16542390, 24357598, 26178382).

It should be noted that acquired (somatic) variants may arise in the blood, particularly with advancing age, and may even appear heterozygous (PMID: 25426837, 25426838, 33864022). Results from blood or saliva specimens alone may not distinguish between germline and acquired variants in this gene and additional testing may be appropriate.

- Biological relatives have a chance of being at risk for NF1-related conditions and should consider testing if clinically appropriate.

Variant details

NF1, Intron 37, c.5546+5G>A (Intronic), heterozygous, PATHOGENIC

- This sequence change falls in intron 37 of the NF1 gene. It does not directly change the encoded amino acid sequence of the NF1 protein. It affects a nucleotide within the consensus splice site.
- This variant is not present in population databases (gnomAD no frequency).
- This variant has been observed in individual(s) with clinical features of neurofibromatosis type 1 (PMID: 16944272, 29673180; Invitae).
- This variant is also known as IVS29+5G>A.
- ClinVar contains an entry for this variant (Variation ID: 653133).
- Variants that disrupt the consensus splice site are a relatively common cause of aberrant splicing (PMID: 17576681, 9536098). Studies have shown that this variant alters mRNA splicing and is expected to lead to the loss of protein expression (PMID: 19292874).
- For these reasons, this variant has been classified as Pathogenic.



Genes analyzed

This table represents a complete list of genes analyzed for this individual, including the relevant gene transcript(s). If more than one transcript is listed for a single gene, variants were reported using the first transcript listed unless otherwise indicated in the report. An asterisk (*) indicates that this gene has a limitation. Please see the Limitations section for details. Results are negative unless otherwise indicated in the report. Benign and Likely Benign variants are not included in this report and in specific scenarios variants of uncertain significance in the requisitioned gene(s) may not be included in this report. These variants are available upon request.

GENE	TRANSCRIPT
NF1*	NM_000267.3

Methods

- Genomic DNA obtained from the submitted sample is enriched for targeted regions using a hybridization-based protocol, and sequenced using Illumina technology. Unless otherwise indicated, all targeted regions are sequenced with $\geq 50\times$ depth or are supplemented with additional analysis. Reads are aligned to a reference sequence (GRCh37), and sequence changes are identified and interpreted in the context of a single clinically relevant transcript, indicated in the Genes Analyzed table. Enrichment and analysis focus on the coding sequence of the indicated transcripts, 20bp of flanking intronic sequence, and other specific genomic regions demonstrated to be causative of disease at the time of assay design. Promoters, untranslated regions, and other non-coding regions are not otherwise interrogated. For some genes only targeted loci are analyzed. Exonic deletions and duplications are called using an in-house algorithm that determines copy number at each target by comparing the read depth for each target in the proband sequence with both mean read-depth and read-depth distribution, obtained from a set of clinical samples. Markers across the X and Y chromosomes are analyzed for quality control purposes and may detect deviations from the expected sex chromosome complement. Such deviations may be included in the report in accordance with internal guidelines. Variants are reported according to the Human Genome Variation Society (HGVS) guidelines. Confirmation of the presence and location of reportable variants is performed as needed based on stringent criteria using one of several validated orthogonal approaches (PubMed ID 30610921). Sequencing is performed by Invitae Corporation (1400 16th Street, San Francisco, CA 94103, #05D2040778). Confirmatory sequencing is performed by Invitae Corporation (1400 16th Street, San Francisco, CA 94103, #05D2040778). RNA sequencing is performed by Invitae Corporation (1400 16th Street, San Francisco, CA 94103, #05D2040778).

The following additional analyses are performed if relevant to the requisition. For PMS2 exons 12-15, the reference genome has been modified to force all sequence reads derived from PMS2 and the PMS2CL pseudogene to align to PMS2, and variant calling algorithms are modified to support an expectation of 4 alleles. If a rare SNP or indel variant is identified by this method, both PMS2 and the PMS2CL pseudogene are amplified by long-range PCR and the location of the variant is determined by Pacific Biosciences (PacBio) SMRT sequencing of the relevant exon in both long-range amplicons. If a CNV is identified, MLPA or MLPA-seq is run to confirm the variant. If confirmed, both PMS2 and PMS2CL are amplified by long-range PCR, and the identity of the fixed differences between PMS2 and PMS2CL are sequenced by PacBio from the long-range amplicon to disambiguate the location of the CNV. For C9orf72 repeat expansion testing, hexanucleotide repeat units are detected by repeat-primed PCR (RP-PCR) with fluorescently labeled primers followed by capillary electrophoresis. Interpretation Reference Ranges: Benign (Normal Range): <25 repeat units, Uncertain: 25-30 repeat units, Pathogenic (Full Mutation): ≥ 31 repeat units (PMID: 21944779, 22406228, 23111906, 28689190, 31315673, 33168078, 33575483). A second round of RP-PCR utilizing a non-overlapping set of primers is used to confirm the initial call in the case of suspected allele sizes of 22 or more repeats. For RNA analysis of the genes indicated in the Genes Analyzed table, complementary DNA is synthesized by reverse transcription from RNA derived from a blood specimen and enriched for specific gene sequences using capture hybridization. After high-throughput sequencing using Illumina technology, the output reads are aligned to a reference sequence (genome build GRCh37; custom derivative of the RefSeq transcriptome) to identify the locations of exon junctions through the detection of split reads. The relative usage of exon junctions in a test specimen is assessed quantitatively and compared to the usage seen in control specimens. Abnormal exon junction usage is evaluated as evidence in the Sherlock variant interpretation framework. If an abnormal splicing pattern is predicted based on a DNA variant outside the typical reportable range, as described above, the presence of the variant is confirmed by targeted DNA sequencing.

- A PMID is a unique identifier referring to a published, scientific paper. Search by PMID at <http://www.ncbi.nlm.nih.gov/pubmed>.
- An rsID is a unique identifier referring to a single genomic position, and is used to associate population frequency information with sequence changes at that position. Reported population frequencies are derived from a number of public sites that aggregate data from large-scale population sequencing projects, including ExAC (<http://exac.broadinstitute.org>), gnomAD (<http://gnomad.broadinstitute.org>), and dbSNP (<http://ncbi.nlm.nih.gov/SNP>).
- A MedGen ID is a unique identifier referring to an article in MedGen, NCBI's centralized database of information about genetic disorders and phenotypes. Search by MedGen ID at <http://www.ncbi.nlm.nih.gov/medgen>. An OMIM number is a unique identifier referring to a comprehensive entry in Online Mendelian Inheritance in Man (OMIM). Search by OMIM number at <http://omim.org/>.
- Invitae uses information from individuals undergoing testing to inform variant interpretation. If "Invitae" is cited as a reference in the variant details this may refer to the individual in this requisition and/or historical internal observations.

Limitations

Based on validation study results, this assay achieves >99% analytical sensitivity and specificity for single nucleotide variants, insertions and deletions <15bp in length, and exon-level deletions and duplications. Invitae's methods also detect insertions and deletions larger than 15bp but smaller than a full exon but sensitivity for these may be marginally reduced. Invitae's deletion/duplication analysis determines copy number at a single exon resolution at virtually all targeted exons. However, in rare situations, single-exon copy number events may not be analyzed due to inherent sequence properties or isolated reduction in data quality. Certain types of variants, such as structural rearrangements (e.g. inversions, gene conversion events, translocations, etc.) or variants embedded in sequence with complex architecture (e.g. short tandem repeats or segmental duplications), may not be detected. Additionally, it may not be possible to fully resolve certain details about variants, such as mosaicism, phasing, or mapping ambiguity. Unless explicitly guaranteed, sequence changes in the promoter, non-coding exons, and other non-coding regions are not covered by this assay. Please consult the test definition on our website for details regarding regions or types of variants that are covered or excluded for this test. This report reflects the analysis of an extracted genomic DNA sample. While this test is intended to reflect the analysis of extracted genomic DNA from a referred patient, in very rare cases the analyzed DNA may not represent that individual's constitutional genome, such as in the case of a circulating hematolymphoid neoplasm, bone marrow transplant, blood transfusion, chimerism, culture artifact or maternal cell contamination. Interpretations are made on the assumption that any clinical information provided, including specimen identity, is accurate. Invitae's RNA analysis is not designed for use as a stand-alone diagnostic method and cannot determine absolute RNA levels. Results from the RNA analysis may not be informative for interpreting copy number gains.

NF1: Sequencing analysis for exons 2, 7, 25, 41, 48 includes only cds +/- 10 bp.

Disclaimer

DNA studies do not constitute a definitive test for the selected condition(s) in all individuals. It should be realized that there are possible sources of error. Errors can result from trace contamination, rare technical errors, rare genetic variants that interfere with analysis, recent scientific developments, and alternative classification systems. This test should be one of many aspects used by the healthcare provider to help with a diagnosis and treatment plan, but it is not a diagnosis itself. This test was developed and its performance characteristics determined by Invitae. It has not been cleared or approved by the FDA. The laboratory is regulated under the Clinical Laboratory Improvement Act (CLIA) as qualified to perform high-complexity clinical tests (CLIA ID: 05D2040778). This test is used for clinical purposes. It should not be regarded as investigational or for research.

This report has been reviewed and approved by:



Andrea Behlmann, PhD, FACMG
Clinical Cytogeneticist & Clinical Molecular Geneticist

This document is not part of Invitae's clinical report and does not represent medical advice. These are general guidelines that are not specific to your result and may not represent all relevant international recommendations. You can use this guide to talk to your healthcare provider about your test results, clinical history, and the most current guidelines. This guide may not be appropriate for results that are suspected to be blood-limited, possibly mosaic, or suggestive of a larger imbalance of genetic material. Invitae recognizes that individuals have diverse gender and sexual identities. In this guide, the terms female, male, women, and men refer to sex assigned at birth.

What is a positive NF1 result?



A positive test result means that a genetic change (variant) was found in the NF1 gene. A positive NF1 variant is considered “pathogenic” or “likely pathogenic” because it is associated with Neurofibromatosis type 1 and NF1-related conditions.

What does this mean?

Neurofibromatosis type 1 is a condition that affects many systems in the body and is associated with an increased risk to develop several types of cancers and tumors. Individuals may have patches of coffee-colored skin (café-au-lait spots), freckles in the underarms and groin areas (inguinal freckling), harmless bumps in the eyes (Lisch nodules), and noncancerous tumors located on or just under the skin (neurofibromas). There is also an increased chance for certain cancers including a type of eye cancer (optic nerve glioma), cancer of the tissues that cover the nerves (malignant peripheral nerve sheath tumors or MPNST), brain tumors, breast cancer, digestive tract tumors (gastrointestinal stromal tumors or GIST) and adrenal gland tumors (pheochromocytoma or PCC). They may also have conditions affecting the heart and blood vessels. Variations of this condition include neurofibromatosis-Noonan syndrome (NFNS) and Watson syndrome, which have similar features as neurofibromatosis type 1 as well as short stature, heart defects, and other symptoms. Symptoms, severity, and age of onset can vary. Variants in NF1 can be mosaic, meaning they are present in some of an individual's cells but not others. Individuals with mosaic NF1 may have more mild symptoms, or even features of the condition that are limited to certain areas of the body. Sometimes an NF1 variant is acquired later in life within the blood, which is known as clonal hematopoiesis. The likelihood of clonal hematopoiesis increases with an individual's age, exposure to chemotherapy, or radiation treatment. Management will differ depending on whether the variant was present at birth or acquired in the blood later on.

What does this mean for family members?

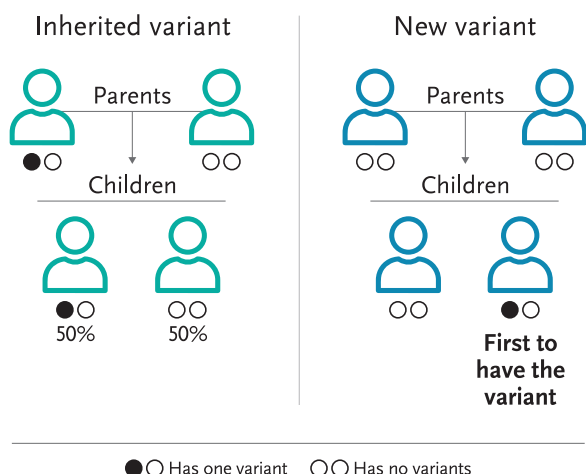


Relatives should be informed about these results. It is recommended that family members talk with their own healthcare provider about a plan for genetic testing and/or health screening. Genetic testing is a personal choice and some individuals may choose not to have genetic testing. Laws protecting employment and health insurance may apply to individuals undergoing genetic testing (for example, the Genetic Information Nondiscrimination Act in the United States).

Will family members have the same variant(s)?

The image shows where a NF1 variant may have come from. Any individual can inherit and pass on a NF1 variant, regardless of sex.

NF1 variants can be inherited from a parent or an individual may be the first person in the family to have a new NF1 variant. Genetic testing of parents may help determine the risk to the individual's siblings and other relatives. Individuals with an NF1 variant present from birth can pass it on to children.



For individuals who are planning a family, reproductive options may be available to help lower the chance of passing on a variant to children.

Create a plan with a healthcare provider



These options are a guide for an individual and their healthcare provider. They are meant to be used along with an individual's genetic test results and other health information as part of a discussion to make a personalized care plan. Each option may or may not be right for an individual. A positive test result on its own cannot predict how a condition may affect an individual. This guide may not be appropriate for results that are suspected to be blood-limited, possibly mosaic, or suggestive of a larger imbalance of genetic material.

Options to consider

TOPIC	OPTION	MORE INFORMATION
Clinical assessment	- Assess whether the NF1 variant is more likely to have been present from birth (which causes neurofibromatosis type 1) or acquired later in the blood (clonal hematopoiesis). - Consider the presence or absence of features of neurofibromatosis type 1, the individual's age, and history of chemotherapy, radiation treatment, or a blood disorder to help determine the most likely explanation. - This guide provides resources for management of individuals diagnosed with neurofibromatosis type 1.	- There are criteria to establish a diagnosis of neurofibromatosis type 1 (PMID: 34012067). - If the underlying reason for the result is not clear, consider additional steps to clarify the nature of this variant. - If the original sample type was blood, saliva, or cheek swab, consider testing DNA from a different sample type, such as a skin biopsy. - Consider testing family members. - If the variant is found in a different sample type or in family members, it suggests neurofibromatosis type 1. However, the extent of involvement in other tissues may still be unclear.
General NF1 management	- Medical evaluation every year (1,2,3) - Whenever possible, individuals with NF1 should be referred to an NF1 specialist for evaluation and care coordination. (1,3,4)	- In childhood and adolescence, evaluation should include measurement of head circumference, growth rate, bone or scoliosis exams, and monitoring for early-onset puberty. (3) - Specialists can be located through the Children's Tumor Foundation (www.ctf.org)
Cutaneous neurofibromas	- Skin exam every year until young adulthood, and then as needed (2,3) - Removal of lesions by a specialist can be considered (1,3)	Cutaneous neurofibromas are noncancerous (benign) tumors, but can impact quality of life due to symptoms or cosmetic appearance.
Plexiform neurofibromas (PN)/ Malignant peripheral nerve sheath tumors (MPNST)	- Assess for the presence of PN at diagnosis. Standard evaluation includes history, physical, and neurological exam. (4) - Imaging can be considered to assess the extent of PN. (1,3,4) - Treatment for PN may include surgery or medication. (4,5) - Education on the signs and symptoms of MPNST (1,2,3) - Imaging is recommended for individuals reporting signs or symptoms consistent with malignant transformation of a PN. (1,3,4)	- Clinical judgment should be used to determine whether asymptomatic PN should be followed with imaging. (1,3) - Currently, there is no consensus on appropriate imaging intervals for monitoring known PN. (4) - PN are noncancerous tumors that have the potential to transform into cancerous MPNST. - Symptoms of MPNST include pain, rapid growth, and changes in consistency. (1,2,3) - Referral to specialized, high-volume centers for MPNST diagnosis and management is recommended. (1,3)

TOPIC	OPTION	MORE INFORMATION
Optic nerve gliomas	Eye exam at least every year from birth to age 8, then every 1-2 years from age 8 to 20 (2)	- Ideally children would be followed by a pediatric ophthalmologist with expertise in the condition. (3) - If diagnosed with an optic nerve glioma, refer to a specialized multidisciplinary team. (2)
Brain tumors and neurologic conditions	- Neurologic exam every year or more frequently if indicated (3) - Imaging is recommended for individuals with symptoms. (2,3)	Symptoms include seizures, severe headaches/intracranial pressure, precocious puberty, encephalopathy, and visual disturbance (1,3)
Cardiovascular conditions	Clinical exam in infancy or at diagnosis to assess for heart defects (3)	Individuals with a heart murmur or certain NF1 variants (microdeletions) should be evaluated with a cardiology examination and echocardiography imaging. (3)
	- Blood pressure monitoring at least once per year beginning in early childhood (1,3) - Investigate and treat the underlying cause of high blood pressure (1,3)	Common causes of high blood pressure include essential hypertension, renal artery stenosis (especially in children and young adults), and pheochromocytoma. (1,3)
Pheochromocytoma (PCC)	- Individuals with symptoms of PCC should undergo screening blood and urine tests. (1) - Testing may include plasma-free metanephrines and/or 24-hour urine test for fractionated metanephrines. (1)	- Symptoms of PCC may include unexplained high blood pressure, sweating, headaches, heart palpitations, paleness, shortness of breath and/or feelings of anxiety or panic. - Screening blood and urine tests, and imaging for PCC is not recommended for individuals without symptoms. (1)
Breast cancer	Mammogram every year starting at age 30 and consider breast MRI with contrast every year from ages 30-50, or 5-10 years before the earliest known breast cancer in the family (whichever comes first) (6)	- Consider the possibility of false-positive MRI results due to presence of breast neurofibromas. (1,6) - There are no data to suggest an increased breast cancer risk after age 50. (6)
	There is insufficient evidence to recommend risk-reducing bilateral mastectomy (surgery to remove the breasts) for all NF1-positive women, but it may be considered based on personal and family history. (6)	
Gastrointestinal stromal tumors (GIST)	There is insufficient evidence to recommend special screening for less common tumors (including GIST) in individuals without symptoms. (1)	- GIST are tumors of the digestive system. They are noncancerous, but they can cause symptoms such as intestinal bleeding or abdominal pain. (3) - GIST are rare in childhood. (3)
Developmental and neuropsychological differences	- Developmental and psychosocial evaluations beginning in infancy (3) - Offer testing (neuropsychological and/or psychoeducation) if there are academic or social concerns. (3) - Screen for depression in adults. (1) - Refer to specialists and support groups as needed. (1,3)	- Infants and toddlers may have mild motor delays. (3) - Learning, speech, and social difficulties are more common in individuals with this condition. (3)
Pregnancy considerations	Pregnant individuals should be monitored more closely because there may be increased risk of high blood pressure and other complications. (1)	Consider referral to a high-risk provider. (1)
Family planning	- Discuss reproductive risks. - Individuals with a NF1 variant have a 50% chance to pass on the variant to a child.	Preconception and prenatal reproductive options are available and could be discussed in more detail with a reproductive specialist.

TOPIC	OPTION	MORE INFORMATION
NF1-related research	- Consider options for clinical trials and/or other research. - To find up-to-date information about available clinical trials, visit ClinicalTrials.gov	There are many factors when considering participation in research studies, including time, location, and whether or not an individual meets the specific requirements to be in a study.

These options include recommendations from PMID: 30006586 (1), PMID: 28620004 (2), PMID: 31010905 (3), PMID: 35657359 (4), PMID: 35788692 (5), and NCCN (6). Referenced with permission from the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Genetic/Familial High-Risk Assessment: Breast, Ovarian, and Pancreatic Version 2.2023 (6). © National Comprehensive Cancer Network, Inc. 2023. All rights reserved. Accessed 01/19/2023. To view the most recent and complete version of the guideline, go online to [NCCN.org](https://www.nccn.org). NCCN makes no warranties of any kind whatsoever regarding their content, use or application and disclaims any responsibility for their application or use in any way. More information about genetics and disease continues to be available, so please always refer to the current guidelines and recommendations when considering surveillance and treatment options. Information on this document may not include all relevant international recommendations and acts as a supplement to the Invitae result report. This information is not meant to replace a discussion with an individual's healthcare provider and should not be considered or interpreted as medical advice. Additional resources provided within this document do not indicate or imply any endorsement by Invitae with respect to any third party or any website or the products or services offered by any third party.

Resources



Genetic counseling can help individuals understand their genetic test results and options for next steps. Reviewing test results with a genetic counselor or other healthcare provider is recommended.

Local or telehealth genetic counselors can be identified using the

Find a Genetic Counselor search tool at nsgc.org (US and Canada). Individuals who had genetic testing through Invitae can also log in to their patient portal (invitae.com) to view their results, contact a genetic counselor, or join Invitae's Patient Insights Network (PIN), an online platform where individuals can share information about their health and experiences to help advance research and drug development.

Connect with advocacy groups and other resources

- Children's Tumor Foundation: www.ctf.org
- Neurofibromatosis Network: www.nfnetwork.org

Notes for personalized assessment